

Cambridge IGCSE™

ADDITIONAL MATHEMATICS**0606/12**

Paper 1

February/March 2026

MARK SCHEME

Maximum Mark: 80

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the February/March 2026 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **16** printed pages.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Mathematics-Specific Marking Principles

- 1 Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
- 2 Unless specified in the question, non-integer answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
- 3 Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
- 4 Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
- 5 Where a candidate has misread a number or sign in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 A or B mark for the misread.
- 6 Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	More information required
A0	Accuracy mark awarded zero
A1	Accuracy mark awarded one
A2	Accuracy mark awarded two
A3	Accuracy mark awarded three
B0	Independent mark awarded zero
B1	Independent mark awarded one
B2	Independent mark awarded two
B3	Independent mark awarded three
BOD	Benefit of the doubt
C	Communication mark
	Incorrect
FT	Follow through
Highlighter	Highlight a key point in the working
ISW	Ignore subsequent work
M0	Method mark awarded zero
M1	Method mark awarded one
M2	Method mark awarded two

Annotation	Meaning
M3	Method mark awarded three
MR	Misread
O	Omission
Off-page comment	Allows comments to be entered at the bottom of the RM marking window and then displayed when the associated question item is navigated to.
On-page comment	Allows comments to be entered in speech bubbles on the candidate response.
Pre	Premature rounding/approximation
SC	Special case
SEEN	Indicates that work/page has been seen
TE	Transcription error
	Correct
XP	Correct answer from incorrect working

MARK SCHEME NOTES

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

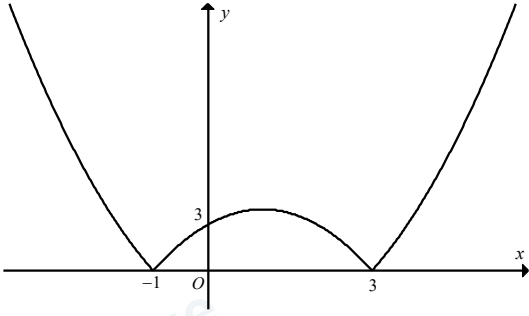
Types of mark

- M Method marks, awarded for a valid method applied to the problem.
- A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. For accuracy marks to be given, the associated Method mark must be earned or implied.
- B Mark for a correct result or statement independent of Method marks.

When a part of a question has two or more ‘method’ steps, the M marks are in principle independent unless the scheme specifically says otherwise; and similarly where there are several B marks allocated. The notation ‘dep’ is used to indicate that a particular M or B mark is dependent on an earlier mark in the scheme.

Abbreviations

awrt	answers which round to
cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
nfw	not from wrong working
oe	or equivalent
rot	rounded or truncated
SC	Special Case
soi	seen or implied

Question	Answer	Marks	Partial marks
1(a)	Shows the remainder to be 0 e.g. $p(-2) = 4(-2)^3 + 8(-2)^2 - (-2) - 2 = 0$ oe	B1	Whatever is shown must be fully correct; must have brackets if needed; for long or synthetic division, remainder of 0 must be indicated
1(b)	$[x + 2](4x^2 - 1) [= 0]$	B1	
	$x^2 = \frac{1}{4}$ or $[x + 2](2x + 1)(2x - 1) [= 0]$	M1	
	$x = -2, \pm \frac{1}{2}$	A1	
2(a)	Correct smooth curve with intercepts indicated 	4	B2 for fully correct smooth curve in correct position with: • A cusp either side of y-axis; • Maximum point in the first quadrant; • Curved middle section; • Correct curvature at outer branches or B1 for correct smooth curve with three of these four properties B1 for y -intercept at 3 indicated; must have attempted correct shape B1 for roots indicated at -1 and 3 ; must have attempted correct shape
2(b)	$2 + \ln \frac{x+1}{4}$ oe, mark final answer	2	M1 for complete method including swapping the variable and changing the subject
3(a)	Straight line from $(0, 0)$ to $(4, 3)$	B2	B1 for straight line $(0, 0)$ to $(4, k)$ or for $(4, 3)$ oe seen
3(b)	$\frac{3}{4} [\text{ms}^{-2}]$ oe	B1	FT $\frac{\text{their } k}{4}$ providing a straight line attempted in (a)
4	Correct simplified equation e.g. $4a - 12 = 10a - 90$ soi	M2	M1 for $\frac{4}{2}(2a + 3(-2))$ or $\frac{10}{2}(2a + 9(-2))$ oe, soi; or for $4a + 6d = 10a + 45d$ oe
	$a = 13$	A1	

Question	Answer	Marks	Partial marks
5	$\frac{dy}{dx} = \frac{2x(3-x) - (x^2-8)(-1)}{(3-x)^2}$ oe, isw OR $\frac{dy}{dx} = 2x(3-x)^{-1} + (x^2-8)(-(3-x)^{-2} \times -1)$ oe, isw	2	M1 for attempt quotient rule or product rule on $(x^2-8)(3-x)^{-1}$
	$x^2 - 6x + 8 = 0$ or $-x^2 + 6x - 8 = 0$	M1	FT <i>their</i> $\frac{dy}{dx} = 0$ providing reduces to a 3-term quadratic
	Solves or factorises <i>their</i> 3-term quadratic e.g. $(x-4)(x-2) [= 0]$ or $(4-x)(x-2) [= 0]$	M1	
	$x = 4, 2$	A1	dep on evidence of a correct derivative
	$y = -8, -4$	A1	dep on evidence of a correct derivative
	$\sqrt{(4-2)^2 + (-8+4)^2} = \sqrt{20}$ isw or $2\sqrt{5}$	A1	FT <i>their</i> x -values and <i>their</i> y -values providing the distance formula calculation is seen
6(a)	$3x + 2 = \pm 7$ oe	M1	
	$x = \frac{5}{3}, x = -3$ mark final answer	A2	A1 for each
	Alternative method		
	$(3x+2)^2 = 49$ oe	(M1)	
	$9x^2 + 12x - 45 [= 0]$ oe	(A1)	
	$x = \frac{5}{3}, x = -3$ mark final answer	(A1)	

Question	Answer	Marks	Partial marks
6(b)	$2x - 4 * x + 1$ and $2x - 4 * -x - 1$ oe	M1	where * is = or any inequality sign
	Critical values $x = 5, x = 1$ soi	A2	A1 for each
	$x > 5, x < 1$ mark final answer	A1	
	Alternative method		
	$(2x - 4)^2 * (x + 1)^2$	(M1)	where * is = or any inequality sign
	$3x^2 - 18x + 15$ [*0] oe	(A1)	
	$x = 5, 1$ soi	(A1)	
	$x < 1, x > 5$ mark final answer	(A1)	
7	$\frac{dy}{dx} : \frac{3}{2}x^{-\frac{1}{2}} - \frac{1}{2}x^{-\frac{3}{2}}$ oe, isw	B2	B1 for each term
	$\frac{\delta y}{h} = \frac{3}{2} \times 4^{-\frac{1}{2}} - \frac{1}{2} \times 4^{-\frac{3}{2}}$ oe	M1	FT their $\frac{dy}{dx}$
	$\frac{11}{16}h$ oe isw	A1	
8(a)	$\frac{dy}{dx} = 2xe^{2x+1} + e^{2x+1}$ oe, isw	3	B1 for $\frac{d}{dx}(e^{2x+1}) = 2e^{2x+1}$ soi M1 for correct structure of product rule, u and v must be correct but du and dv do not need to be correct
8(b)	$\frac{dy}{dx}\bigg _{x=\frac{1}{2}} = 2e^2$	M1	FT their $\frac{dy}{dx}\bigg _{x=\frac{1}{2}}$ providing their $\frac{dy}{dx}$ is a sum of terms oe
	When $x = \frac{1}{2}, y = \frac{1}{2}e^2$	B1	
	$y - \frac{1}{2}e^2 = 2e^2\left(x - \frac{1}{2}\right)$ oe	M1	FT their $\frac{dy}{dx}\bigg _{x=\frac{1}{2}}$ and $\left(\frac{1}{2}, \text{their } y\right)$
	$\left(0, -\frac{1}{2}e^2\right), \left(\frac{1}{4}, 0\right)$	A2	A1 for each or for finding the x -intercept as $\frac{1}{4}$ and the y -intercept as $-\frac{1}{2}e^2$

Question	Answer	Marks	Partial marks
8(c)	$\frac{1}{2}\left(xe^{2x+1} - \frac{1}{2}e^{2x+1}\right) + c$ oe	B4	B3 for $\frac{x}{2}e^{2x+1} - \frac{1}{4}e^{2x+1}$ oe or for $\int 2xe^{2x+1} dx = xe^{2x+1} - \frac{1}{2}e^{2x+1} + c$ or for $\int xe^{2x+1} dx = \frac{x}{2}e^{2x+1} + ke^{2x+1} + c$ where $k = \frac{1}{4}$ or $k = -\frac{1}{2}$ or for $\int xe^{2x+1} dx = mxe^{2x+1} - \frac{1}{4}e^{2x+1} + c$ where $m = 1$ or $m = \frac{1}{4}$ or $m = -\frac{1}{2}$ or B2 for $\int xe^{2x+1} dx = \frac{x}{2}e^{2x+1} - \int \frac{1}{2}e^{2x+1} dx$ or for $\frac{1}{2}e^{2x+1} + \int 2xe^{2x+1} dx = xe^{2x+1}$ or for $\int 2xe^{2x+1} dx = xe^{2x+1} - \frac{1}{2}e^{2x+1}$ or for $\int xe^{2x+1} dx = \frac{x}{2}e^{2x+1} + ke^{2x+1}$ where $k = \frac{1}{4}$ or $k = -\frac{1}{2}$ or for $\int xe^{2x+1} dx = mxe^{2x+1} - \frac{1}{4}e^{2x+1}$ where $m = 1$ or $m = \frac{1}{4}$ or $m = -\frac{1}{2}$ or B1 for $\int 2xe^{2x+1} dx = xe^{2x+1} - \int e^{2x+1} dx$ or for $xe^{2x+1} = \int 2xe^{2x+1} dx + \int e^{2x+1} dx$ soi
9(a)(i)	$\begin{pmatrix} -3 \\ 4 \end{pmatrix} + t \begin{pmatrix} 4 \\ -2 \end{pmatrix}$ or $\begin{pmatrix} -3+4t \\ 4-2t \end{pmatrix}$ oe	B1	

Question	Answer	Marks	Partial marks
9(a)(ii)	Position vector of A : $t \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ soi	B1	
	$t \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} -3 \\ 4 \end{pmatrix} + t \begin{pmatrix} 4 \\ -2 \end{pmatrix}$ soi	M1	FT <i>their</i> position vectors in terms of t
	A correct pair of equations e.g. $2t = -3 + 4t$ and $t = 4 - 2t$	M1	OR finds one value of t from one correct equation and uses it in both sides of the vector equation e.g. $2t = -3 + 4t \rightarrow t = \frac{3}{2}$ and $\frac{3}{2} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} -3 \\ 4 \end{pmatrix} + \frac{3}{2} \begin{pmatrix} 4 \\ -2 \end{pmatrix}$
	$t = \frac{3}{2}$ and $t = \frac{4}{3}$ oe and concludes particles do not collide OR finds one value of t and shows it is not consistent with the other equation and concludes particles do not collide	A1	OR finds a correct, inconsistent pair of position vectors e.g. $\begin{pmatrix} 3 \\ 1.5 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ 1 \end{pmatrix}$ and concludes particles do not collide

Question	Answer	Marks	Partial marks
9(b)	$\overrightarrow{QR} = -4\mathbf{i} + 2\mathbf{j}$ oe, soi OR $\sqrt{(3-1)^2 + (2-4)^2}$ oe	B1	
	$ \overrightarrow{QR} = 2\sqrt{5}$ nfw and $S(1, 3)$ oe, soi	B2	B1 for $ \overrightarrow{QR} = 2\sqrt{5}$ nfw
	$\overrightarrow{RS} = -2\mathbf{i} + \mathbf{j}$ oe, soi	B1	
	$a = 1, b = 3$ $a = -3, b = 5$ nfw	B1	
	Alternative method 1: $\overrightarrow{SR} = k\overrightarrow{QR}$		
	$(-4k)^2 + (2k)^2 = 5$ oe	(B2)	B1 for $\overrightarrow{QR} = -4\mathbf{i} + 2\mathbf{j}$ oe soi
	$k = \pm \frac{1}{2}$	(B1)	
	$\frac{1}{2}(-4) = 1 + a$ and $\frac{1}{2}(2) = b - 4$ oe or $-\frac{1}{2}(-4) = 1 + a$ and $-\frac{1}{2}(2) = b - 4$ oe	(B1)	OR $\overrightarrow{OS} = \begin{pmatrix} -1 \\ 4 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} -4 \\ 2 \end{pmatrix}$ oe or $\overrightarrow{OS} = \begin{pmatrix} -1 \\ 4 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} -4 \\ 2 \end{pmatrix}$ oe
	$a = 1, b = 3$ $a = -3, b = 5$ nfw	(B1)	
	Alternative method 2		
	Correct relationship between a and b e.g. $\frac{4-2}{-1-3} = \frac{b-4}{a+1}$ oe or $\frac{4-2}{-1-3} = \frac{b-2}{a-3}$ oe or $(a+1)^2 + (b-4)^2 = 5$	(B2)	B1 for $\overrightarrow{QR} = -4\mathbf{i} + 2\mathbf{j}$ oe soi or $\overrightarrow{RS} = (a+1)\mathbf{i} + (b-4)\mathbf{j}$ oe soi or $\overrightarrow{QS} = (a-3)\mathbf{i} + (b-2)\mathbf{j}$ oe soi or $m_{QR} = \frac{4-2}{-1-3}$ oe or $m_{RS} = \frac{b-4}{a+1}$ oe or $m_{QS} = \frac{b-2}{a-3}$ oe
	Eliminates a or b e.g. $(8-2b)^2 + (b-4)^2 = 5$ or $(a+1)^2 + \left(\frac{7-a}{2} - 4\right)^2 = 5$	(B1)	
	$5b^2 - 40b + 75 [= 0]$ oe or $5a^2 + 10a - 15 [= 0]$ oe	(B1)	
	$a = 1, b = 3$ $a = -3, b = 5$ nfw	(B1)	

Question	Answer	Marks	Partial marks
10(a)	$\log_5 x - \frac{3}{\log_5 x} = 2$ or $\frac{1}{\log_x 5} - 3\log_x 5 = 2$	M2	M1 for a correct change of base resulting in logarithms with a consistent base in a correct equation
	$(\log_5 x)^2 - 2\log_5 x - 3 [= 0]$ or $3(\log_x 5)^2 + 2\log_x 5 - 1 [= 0]$	A1	Note: A0 for $\log_5 x^2 - 2\log_5 x - 3 [= 0]$ or $3\log_x 5^2 + 2\log_x 5 - 1 [= 0]$
	Solves or factorises <i>their</i> 3-term quadratic e.g. $(\log_5 x - 3)(\log_5 x + 1)$ or $(3\log_x 5 - 1)(\log_x 5 + 1)$	M1	dep on a correct change of base and no invalid logarithm work
	$\log_5 x = 3$ $\log_5 x = -1$ or $\log_x 5 = \frac{1}{3}$ $\log_x 5 = -1$	A1	
	$x = 125, \frac{1}{5}$ oe, nfw	A1	
10(b)	$16(2^{3x})^2 - 48 \times 2^{3x} - 64 [= 0]$ oe or $16 \times 8^{2x} - 48 \times 8^x - 64 [= 0]$ oe or $(2^{3x+2})^2 - 12 \times 2^{3x+2} - 64 [= 0]$	B1	
	Solves or factorises <i>their</i> 3-term quadratic in 2^{3x} or 8^x or 2^{3x+2} e.g. $(2^{3x} - 4)(2^{3x} + 1) [= 0]$ or $(8^x - 4)(8^x + 1) [= 0]$ oe	M1	or $(2^{3x+2} - 16)(2^{3x+2} + 4) [= 0]$
	$2^{3x} = 4, [2^{3x} = -1]$ or $8^x = 4, [8^x = -1]$ oe	A1	or $2^{3x+2} = 16, [2^{3x+2} = -4]$
	$x = \frac{2}{3}$ nfw	A1	

Question	Answer	Marks	Partial marks
11	$-(x-a)(x-2a) = 0$ oe or $\frac{3a \pm \sqrt{(3a)^2 - 4[1](2a^2)}}{2}$ oe	M1	
	$x = a$ and $x = 2a$	A1	
	Correct plan $\int_{\text{their } a}^{\text{their } 2a} (-x^2 + 3ax - 2a^2) dx = \frac{9}{16}$ soi	M1	FT <i>their</i> intersections with x -axis providing they are derived from solving $0 = -x^2 + 3ax - 2a^2$ oe
	$\left[-\frac{x^3}{3} + \frac{3ax^2}{2} - 2a^2x \right]_{\text{their } a}^{\text{their } 2a}$	A1	FT <i>their</i> intersections with x -axis providing they are derived from solving $0 = -x^2 + 3ax - 2a^2$ oe
	Correct substitution of <i>their</i> limits into <i>their</i> expression e.g. $-\frac{8}{3}a^3 + 6a^3 - 4a^3 - \left(-\frac{1}{3}a^3 + \frac{3}{2}a^3 - 2a^3 \right) \left[= \frac{9}{16} \right]$	M1	dep previous M1 ; FT <i>their</i> expression of the form $Ax^3 + Bx^2 + Cx$ and <i>their</i> limits providing they are expressions in a
	$a = \frac{3}{2}$	A1	

Question	Answer	Marks	Partial marks
12	Subtracts the equations of the circles to eliminate x^2 and y^2 OR finds the gradient or the midpoint of the common chord using $(-1, 2)$ and $(6, 3)$	M1	condone one sign, arithmetic or miscopy error
	Equation of common chord e.g. $14x + 2y - 40 = 0$ oe, isw OR $y - 2.5 = -7(x - 2.5)$ oe, isw	A1	
	Uses <i>their</i> equation of the common chord to eliminate one unknown	M1	FT <i>their</i> equation of the common chord providing it was found by attempting a valid method
	$x^2 + 2x + (20 - 7x)^2 - 4(20 - 7x) - 20 [= 0]$ or $x^2 - 12x + (20 - 7x)^2 - 6(20 - 7x) + 20 [= 0]$ oe, soi	A1	OR $\left(\frac{20-y}{7}\right)^2 + 2\left(\frac{20-y}{7}\right) + y^2 - 4y - 20 [= 0]$ or $\left(\frac{20-y}{7}\right)^2 - 12\left(\frac{20-y}{7}\right) + y^2 - 6y + 20 [= 0]$ oe, soi
	$50x^2 - 250x + 300 [= 0]$ or $x^2 - 5x + 6 [= 0]$	A1	OR $50y^2 - 250y - 300 [= 0]$ or $y^2 - 5y - 6 [= 0]$
	Solves or factorises <i>their</i> 3-term quadratic in x or y	M1	dep previous M1 FT <i>their</i> 3-term quadratic
	$x = 3, 2$	A1	OR $y = -1, 6$
	$(3, -1) (2, 6)$	A1	Accept $x = \dots y = \dots$ if pairing clearly implied

Question	Answer	Marks	Partial marks
12	Alternative method		
	$(x+1)^2 + (y-2)^2 = 25$ or $(x-6)^2 + (y-3)^2 = 25$ and makes e.g. $(y-2)^2$ or $(x+1)^2$ the subject	M1	condone one sign, arithmetic or miscopy error
	$y = [\pm]\sqrt{25 - (x+1)^2} + 2$ oe, isw or $x = [\pm]\sqrt{25 - (y-2)^2} - 1$ oe, isw	A1	
	Uses <i>their</i> expression for y or x to eliminate one unknown	M1	FT <i>their</i> expression providing it was found by attempting a valid method
	$(x-6)^2 + ([\pm]\sqrt{25 - (x+1)^2} + 2 - 3)^2 = 25$ or $(x+1)^2 + ([\pm]\sqrt{25 - (x-6)^2} + 3 - 2)^2 = 25$ oe	A1	OR $([\pm]\sqrt{25 - (y-2)^2} - 1 - 6)^2 + (y-3)^2 = 25$ or $([\pm]\sqrt{25 - (y-3)^2} + 6 + 1)^2 + (y-2)^2 = 25$ oe
	$50x^2 - 250x + 300 [=0]$ or $x^2 - 5x + 6 [=0]$	A1	OR $50y^2 - 250y - 300 [=0]$ or $y^2 - 5y - 6 [=0]$
	Solves or factorises <i>their</i> 3-term quadratic in x or y	M1	dep previous M1 FT <i>their</i> 3-term quadratic
	$x = 3, 2$	A1	OR $y = -1, 6$
	$(3, -1) (2, 6)$	A1	Accept $x = \dots y = \dots$ if pairing clearly implied